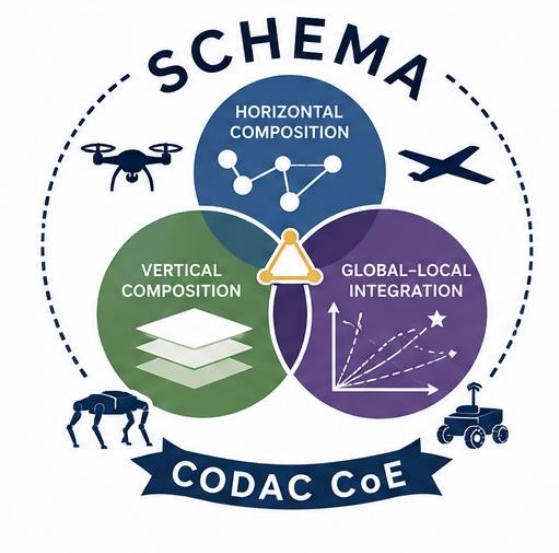


SCHEMA

Scalable Compositional Hierarchies for Evolvable Multi-Agent Architectures

Gioele Zardini

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with Aaron Ames, Ali Jadbabaie, Nikolai Matni, George Pappas, Alberto Sangiovanni-Vincentelli



CODAC University Center of Excellence Kickoff · Vision and Year 1 objectives

24 September 2026 · Van Metre Hall, George Mason University, Arlington VA

MIT · University of Pennsylvania · Caltech · UC Berkeley

schema-coe.github.io

The PIs of SCHEMA



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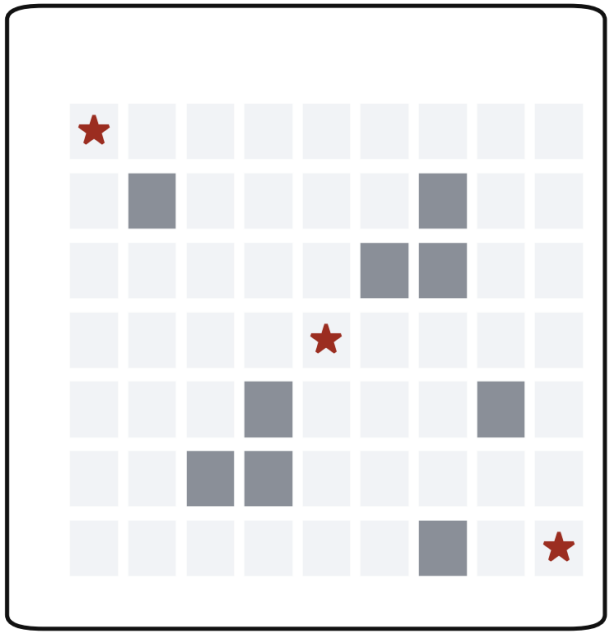
Alberto Sangiovanni-Vincentelli
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Ali Jadbabaie
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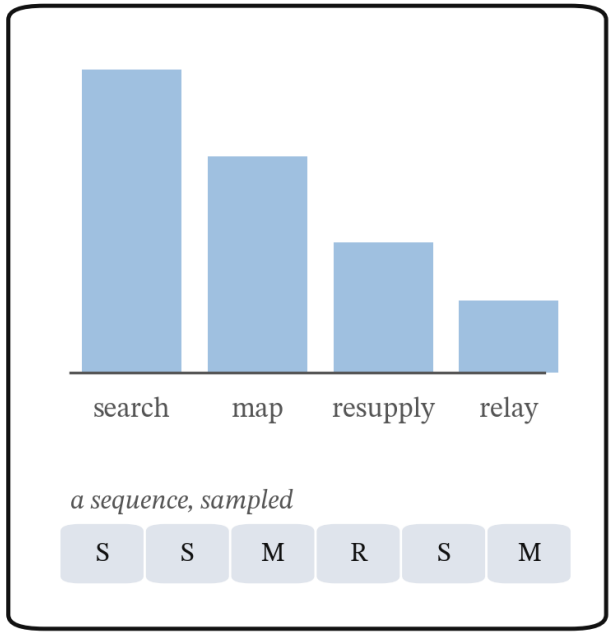
Designing and operating multi-agent systems

an environment



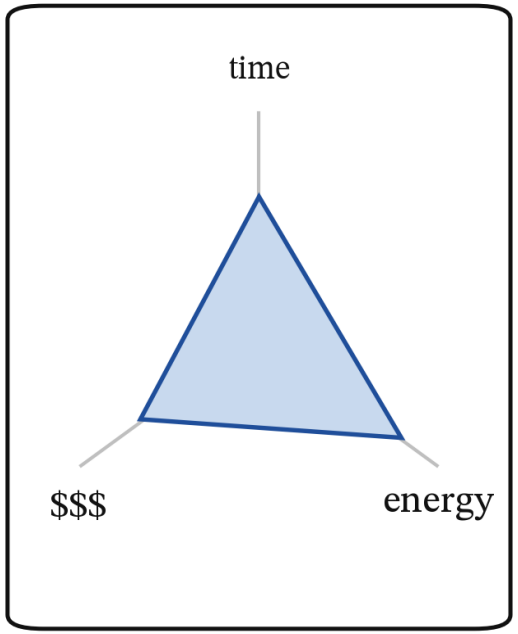
terrain, obstacles, comms shadows,
places that matter

a task, or a distribution of tasks



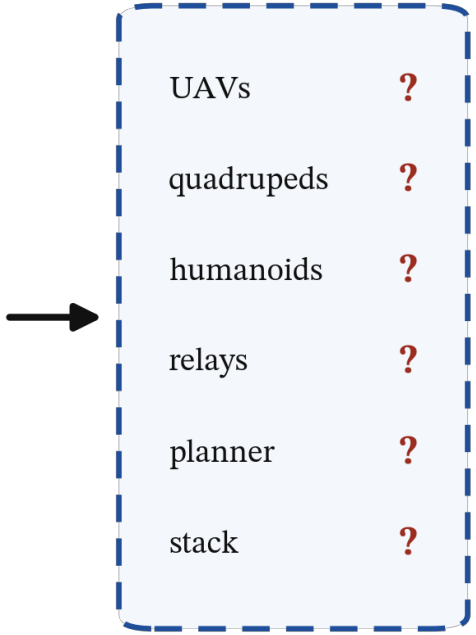
search and rescue, mapping, resupply,
in proportions you do not fully control

three things you pay



and you would rather not
pick which one to lose

the fleet

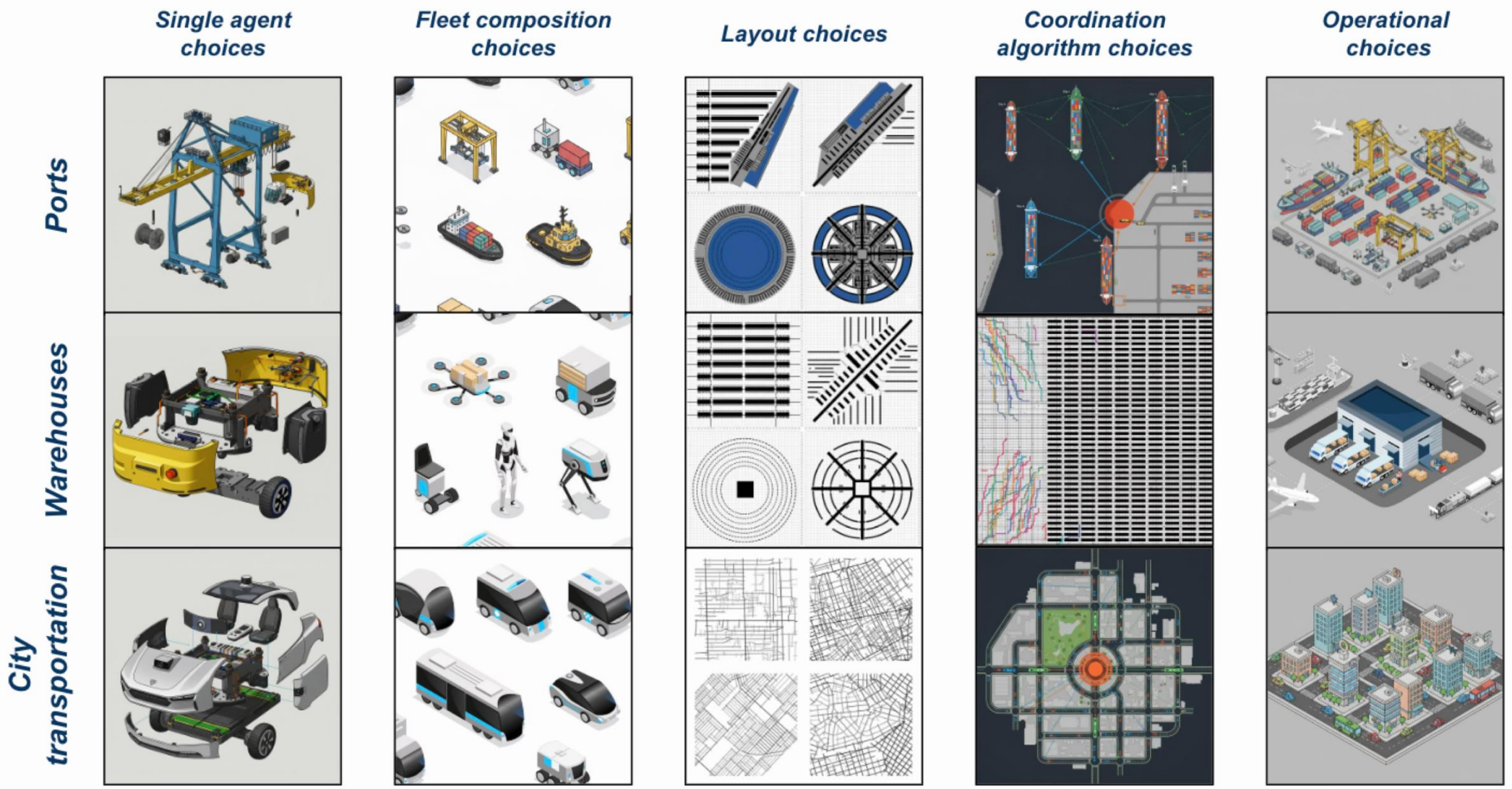


which, how many,
running what

Find me the best fleet!

...where would you start?

The success depends on many components, and their orchestration!



Today each of the five is designed on its own, and the fleet is whatever falls out when they are bolted together.

Multi-agent Autonomy as the frontier of complexity for compositional design

A fleet of autonomous vehicles



=

hardware, software, behavior, coordination, actuation, localization, planning, invasivity, sensing, control, interaction, learning, liability, computation, perception, mapping, regulations, energetics, communication, infrastructure

OMG!

So many components (hardware, software, …), objectives and **choices** to make!

Nobody understands the **whole** thing!

We forget why we made **choices**, and we are afraid to make **changes** (high failure cost).

We need **faster** design cycles, **nimbler** execution.

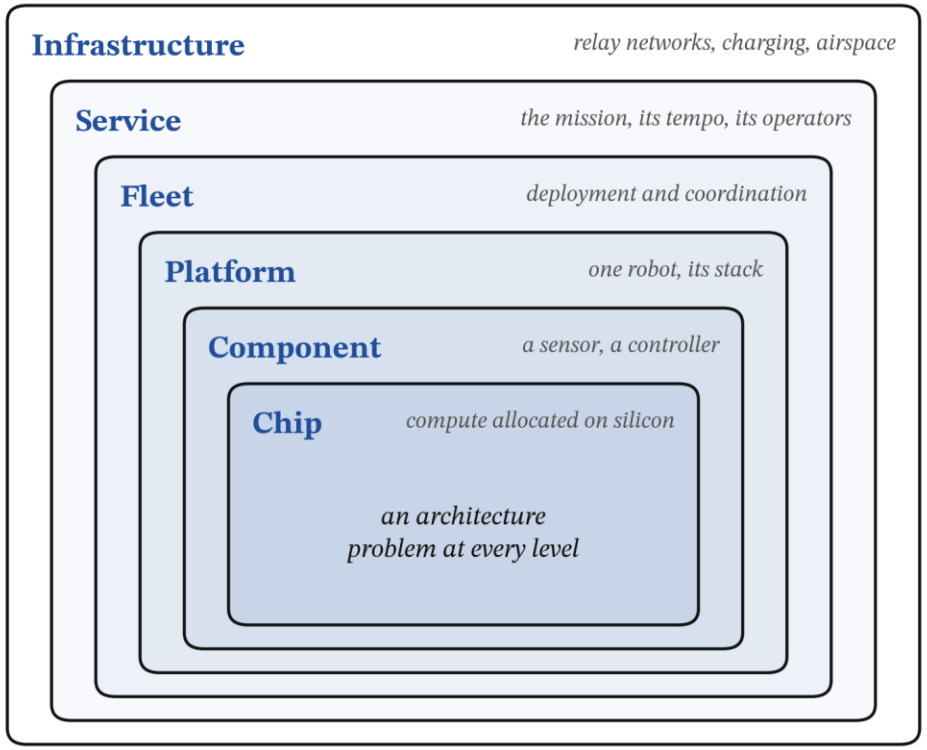
anthropomorphization of 21st century engineering malaise



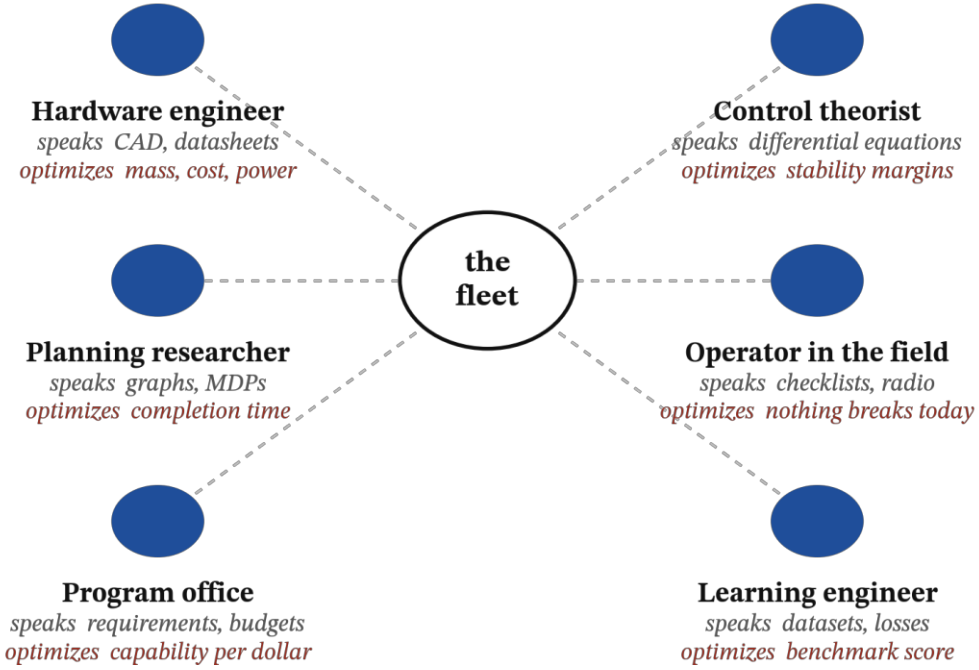
"My dear, it's simple: you lack a *practical* theory of **co-design**!"

Formal,
Quantitative
Intellectually tractable

Your system is just a component in someone else's system



Guarantees have to pass through those boundaries in both directions.

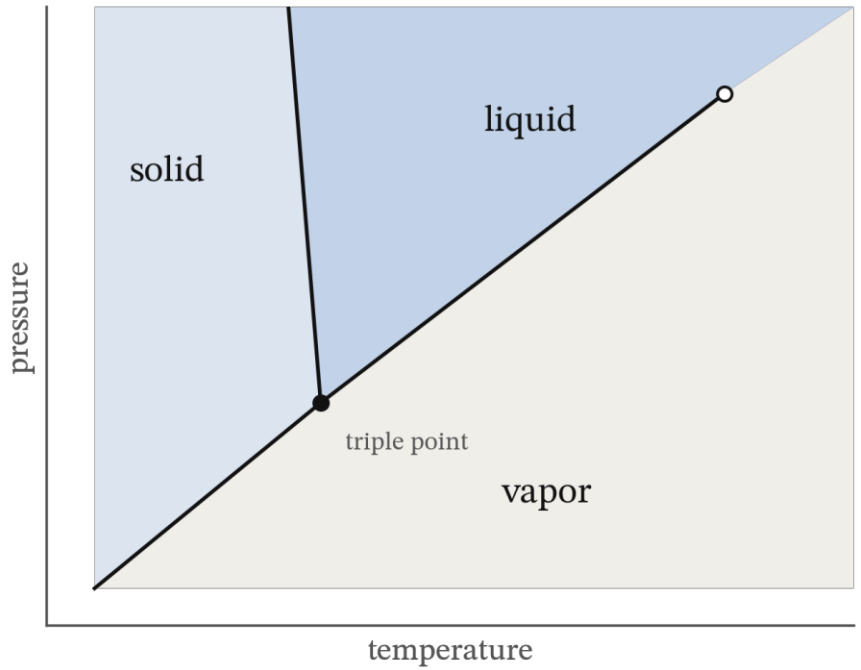


A shared formalism has to let each keep their own model and still compose.

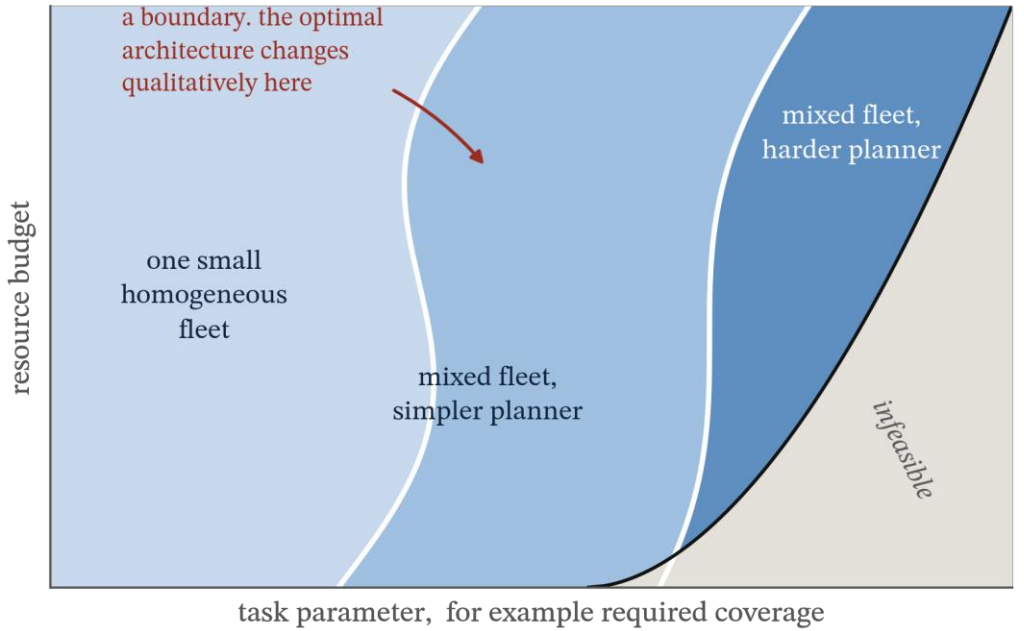
Desiderata for SCHEMA

- **Formal, domain-independent**
- **Computationally tractable**
  - Need to compute solutions efficiently
- **Compositional, hierarchical**
  - My system is a component of somebody else's system
- **Collaborative**
  - Pooling knowledge from experts across fields.
- **Intellectually tractable**
  - Not exclusively accessible to system architects
- **Continuous/adaptive**
  - Design is not static: it should be reactive to changes in goals and contexts

Toward phase diagrams for multi-agent systems design & operation



same idea



Same molecule everywhere. Across a boundary nothing about water changes, only which arrangement wins.

Same robots everywhere. Across a boundary nothing about a robot changes, only which architecture is Pareto optimal.

Agentic AI stacks ask the same question, which architecture, when. Nobody has that map either.

Why we need a new set of mathematical patterns

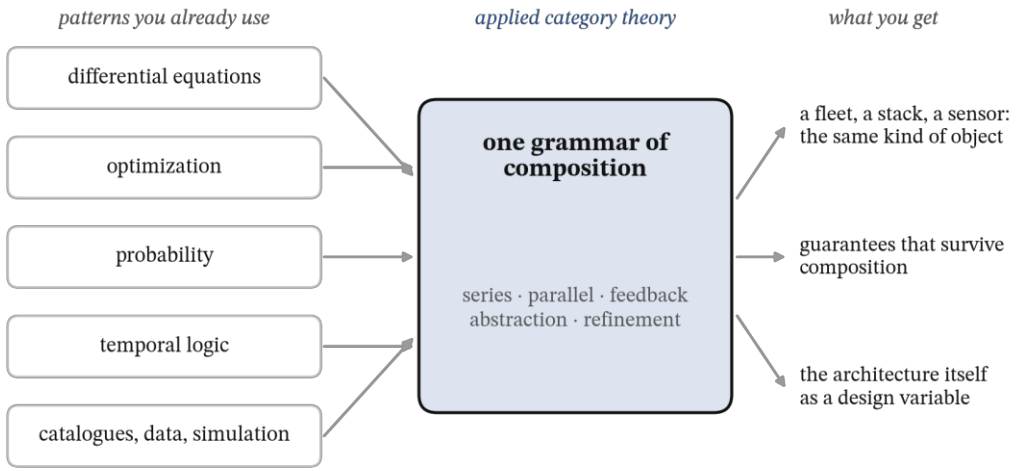
You already have very good mental models of the components and systems in your domain. Some formal, some informal; some computational, some not.

The formal ones do the work. You design, simulate and analyze with them, and you communicate with them (CDC, ICRA).

Mathematics is a set of patterns for describing the world: differential equations, optimization, probability. Patterns built for the problems the engineers before us solved.

Our problems are different. Very complex systems, architectures and interactions. The classical patterns describe the parts; they are silent on how the parts compose.

Applied category theory is an updated set of mathematical patterns, suited to large systems with rich interaction, interconnection and decomposition structure.



Not a replacement for your models. A grammar for how they fit together.

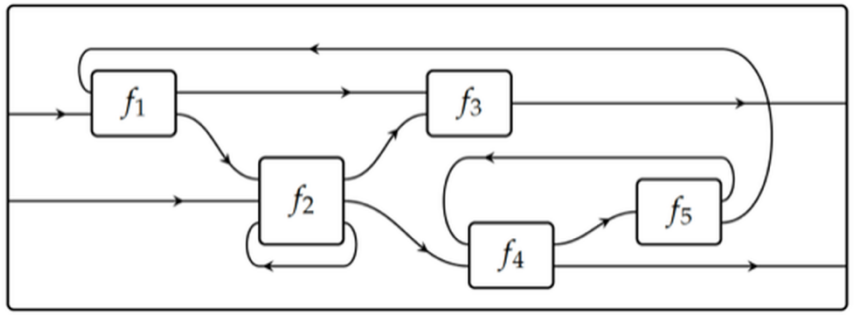
What applied category theory does, and what it does not

A theory with theorems and libraries, and a formal and visual language

WHAT IT DOES

A mathematical theory with theorems, methods and libraries of reusable concepts and connections.

A formal and visual language for reasoning and for design.



The same boxes-and-wires diagram (a traced monoidal category) means wildly different things in different domains: a fleet, a control loop, a supply chain. The composition rules are the same. Study them once, abstractly; re-particularize the conclusions to the domain.

WHAT IT DOES NOT

It does not solve your domain problem. Not by itself, not yet. What it does is help you formulate the problem well, so that the right questions become askable.

The catch. You invest more work at the beginning: definitions, formalization, modeling choices. The problem has to be written down before it can be composed.

The payoff comes later. Things work out. Useful properties are proved cheaply, or come for free, once the structure is right. Closure, tractability and guarantees under composition are theorems, not hopes.

Study composition once. Re-particularize everywhere.

Cases in point: one substrate under everything SCHEMA does

Six places the same grammar shows up, the theme each one feeds, and who is speaking about it today

Co-design

A design problem is a monotone feasibility relation between functionality and resources. Series, parallel and feedback return a design problem: a traced monoidal category. And *solution(composition(a, b)) = composition(solution(a), solution(b))*: the solver is a functor.

Theme A · Zardini, 13:30

Assume-guarantee contracts

Refinement, composition and quotient are the algebra. Contracts are a special case of design problems, so one set of operators serves specification, verification and optimization.

Themes A, C · Sangiovanni-Vincentelli

Categorical Lyapunov theory

Stability stated for the category of flows, not for one model. What is proved for an abstraction transports along the maps that relate the layers.

Theme B · Ames, 12:05

Layered control and abstraction

A reduced-order model is a morphism; bisimulation metrics and trajectory lifting say how good it is. Consistency across an interface becomes a refinement check.

Theme B · Matni, 11:35 · Pappas

Uncertainty, compactly

Markov categories: distributions over design problems compose exactly as design problems do. Parametric uncertainty travels through composition and queries stay functorial.

Themes A, C · Zardini, Jadbabaie

Architecture as a design variable

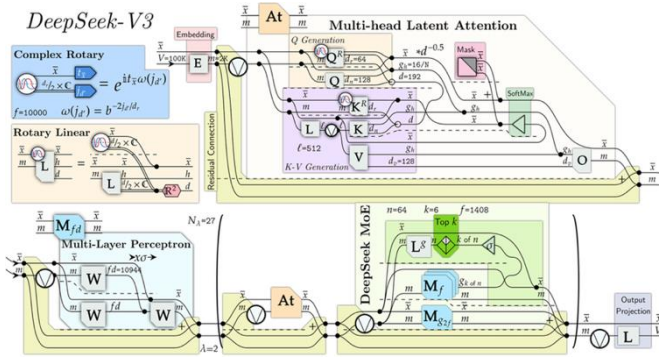
Agents and layers are two dimensions of one meta-model. Compositional Pareto optimality, sweet-spot and coupling metrics, and the phase diagrams all live there.

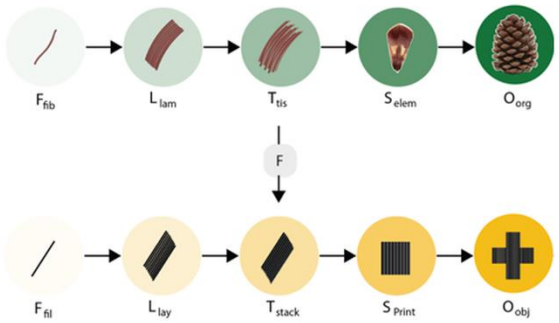
Theme C · all six of us

The same substrate outside autonomy, and how we teach it

Two results that have nothing to do with fleets, and the on-ramp for the people who will build SCHEMA

BEYOND CO-DESIGN





Deep-learning architectures, formalized

DeepSeek-V3 written down as a composite whose every wire carries a type. The diagram is the specification, and the specification is checkable.



Vincent Abbott
TMLR'25, TMLR'26

Nature-derived compositional material design

Fiber to organ, filament to object: two hierarchies and one functor between them, so what nature composes can be re-composed for printing.



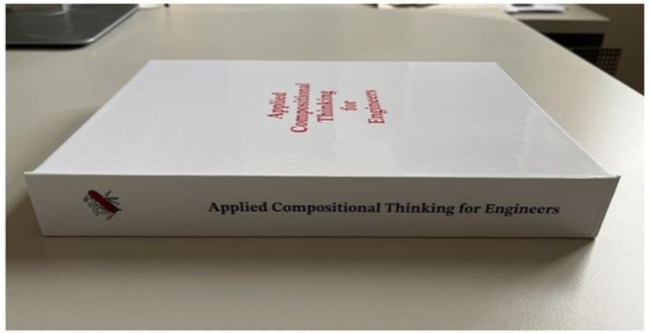
Lee Marom
Markus Bühler

LOWERING THE BARRIER

The math is not hard, it is unfamiliar. Engineering training is heavy on continuous mathematics and light on the discrete structures the theory rests on.

The material is written for someone else. Mathematicians, computer scientists, software engineers. No case studies an engineer recognizes; too broad where we need little, too narrow where we need more.

So we wrote the class and the book. Applied Compositional Thinking for Engineers is the on-ramp for SCHEMA students and for our AFRL colleagues.



Three themes, one formalism, three directions

Theme A

Horizontal composition

Heterogeneous agents and infrastructure composed sideways. Categorical co-design, contracts, uncertainty-aware optimization.

Theme B

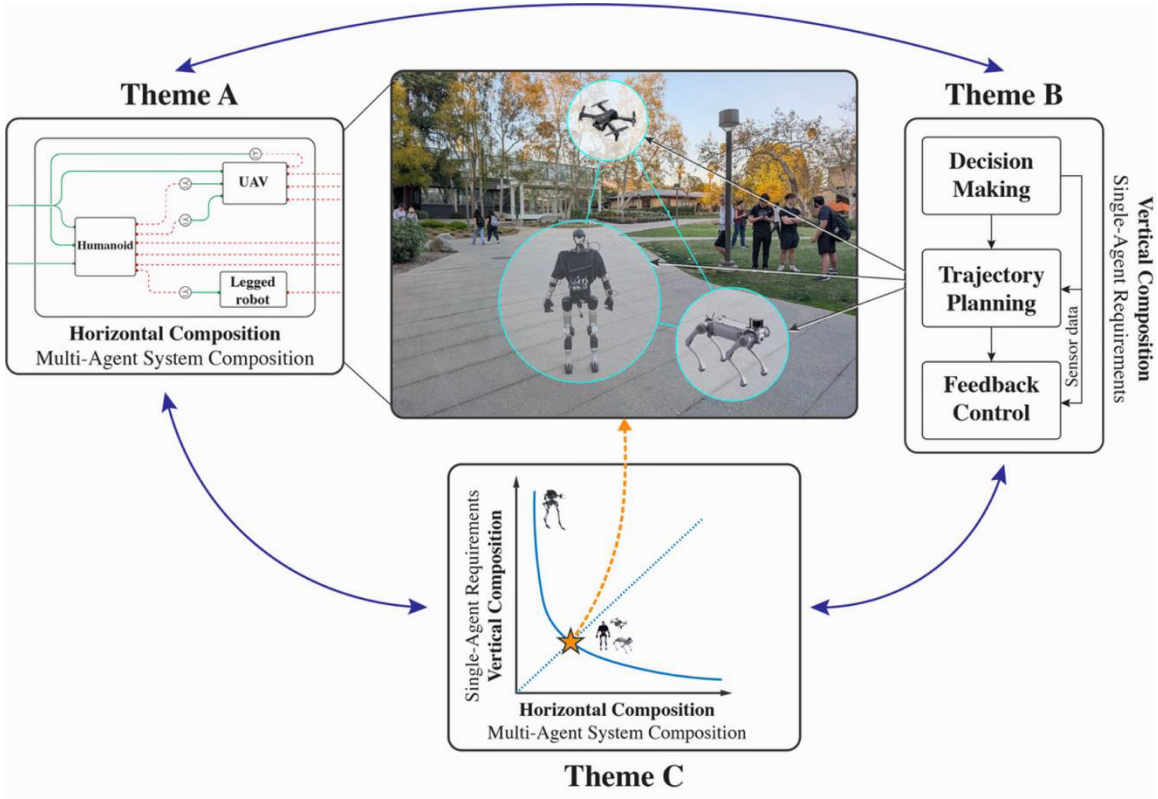
Vertical composition

Objectives that stay feasible and safe when handed to one agent. Layered control with reduced-order models, consistent across layers.

Theme C

Unified composition

Global and local integration in one meta-model. Compositional Pareto optimality across layers and agents, and the metrics for it.



THE FIVE CHALLENGES

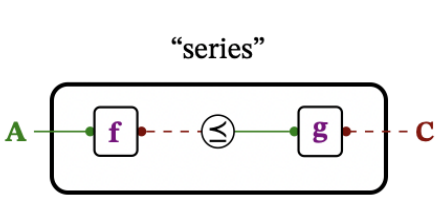
- **C1** Scale
- **C2** Heterogeneity
- **C3** Partial specifications
- **C4** Decentralization
- **C5** Context dependence

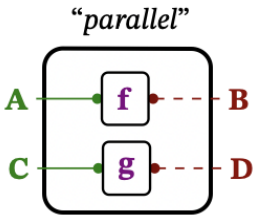
A carries C1, C2 and C3.
B carries C1, C2, C4 and C5.
C spans all five.

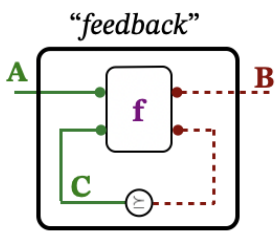
Each theme carries three tasks. X.1 theory, X.2 operationalization, X.3 scalability.

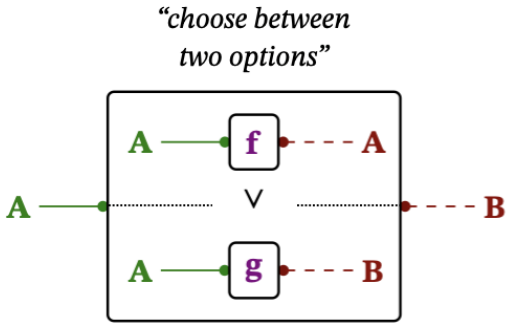
IN CO-DESIGN TERMS a design problem is a monotone relation between functionality and resources, closed under series, parallel and feedback. A wires them sideways, B stacks them, C asks which wiring.

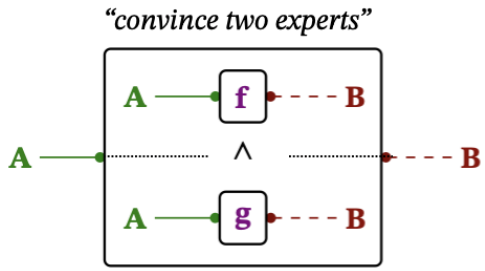
Theme A. Every component becomes the same kind of object

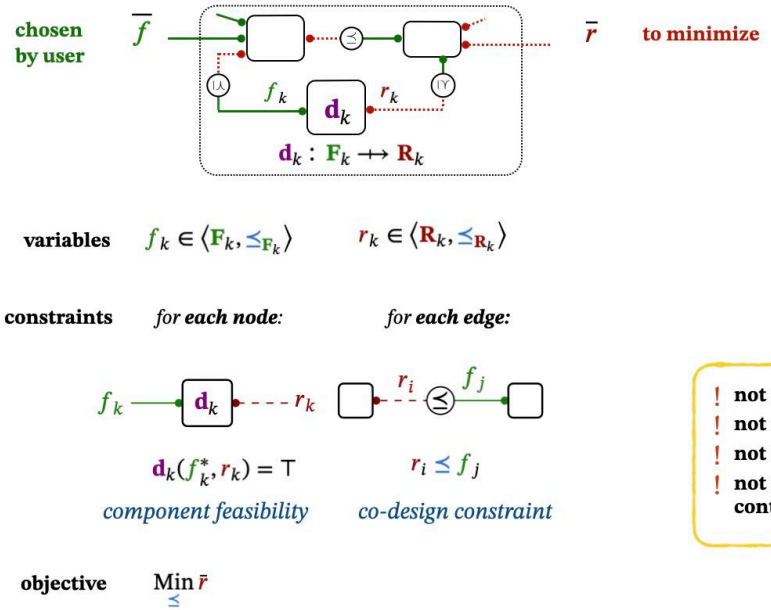












! not convex
! not differentiable
! not continuous
! not even defined on continuous spaces

Populated from anything

Physical models, catalogs, simulation, learned surrogates, or an optimization solved inside the box. Same interface every time, so each expert contributes in their own language.

Closed under composition

Series, parallel and feedback of design problems return a design problem. A fleet, a stack and a sensor are the same kind of object at three scales.

Contracts on the wires

Each component states what it assumes of its environment and what it guarantees in return. Contracts are a special case of design problems, so one algebra serves both.

Theme A. Objectives and tasks

TODAY	WITH SCHEMA
Fleet, planner and infrastructure are modeled by different teams in formalisms that do not talk to each other.	▸ One model. Design problems with contracts on every interface, that any domain expert can populate without adopting anyone else's formalism.
A design is a point. Change the mission or lose a component and you start over.	▸ A design is a Pareto front over the mission. Inverse queries return the minimal local change that restores feasibility.
Uncertainty is handled by margins chosen by hand, and guarantees stop at the component.	▸ Uncertainty carried through composition as bounds and distributions, with guarantees that hold for the composite.

A.1 Compositional modeling and optimization

Design problems and contracts for the mixed fleet, perception to energy to infrastructure, with uncertainty through Markov categories.

Zardini lead · Ames, Matni, Jadbabaie

A.2 Operationalization

Synthesis pipeline from mission specification to fleet. Distributed co-design across stakeholders. Inverse queries. Mission-aware re-tasking under failures.

Zardini, Sangiovanni-Vincentelli, Ames, Matni, Jadbabaie

A.3 Scalability

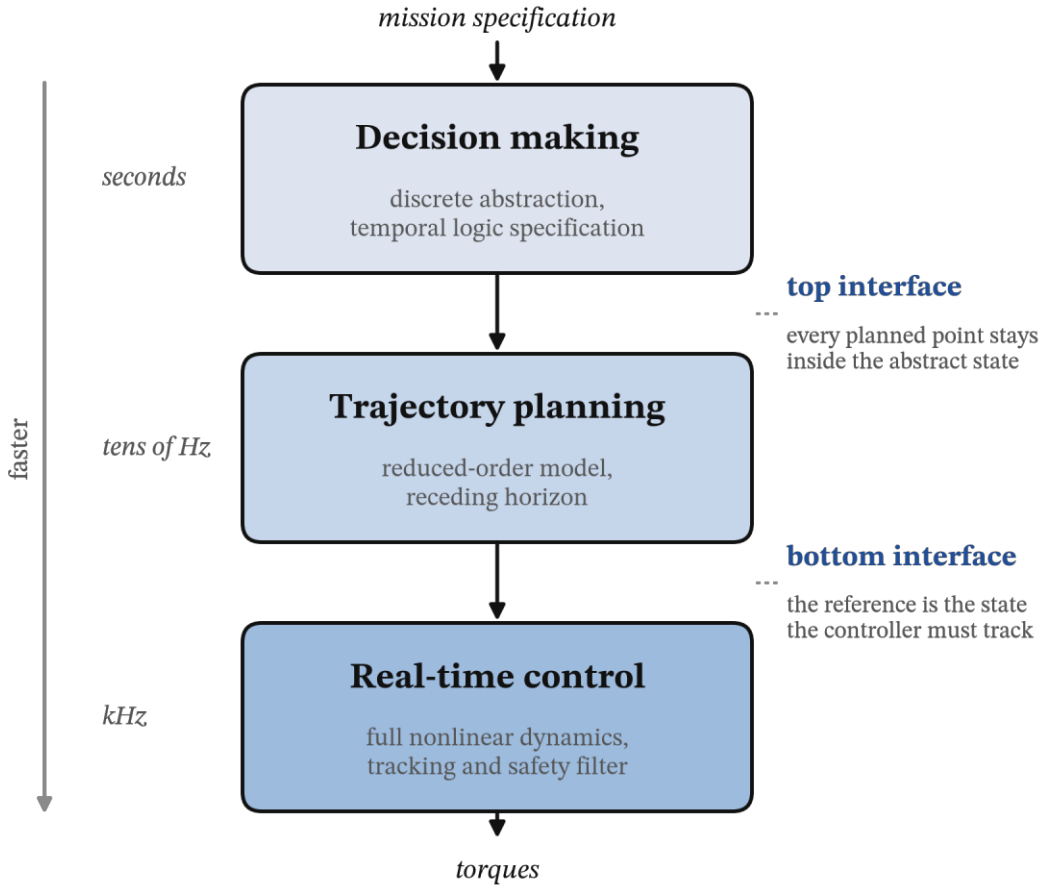
Parallel functorial solving, adaptive sampling of the design space, resource-aware inference, guarantees preserved at fleet scale.

Zardini, Jadbabaie, Matni, Pappas, Ames, Sangiovanni-Vincentelli

Year 1 Task A.1 through the year. Task A.2 starts in the second half. Deliverables, design-problem models of the mixed fleet with contract interfaces, a first uncertainty-aware version, and a working skeleton of the synthesis pipeline.

Theme B. One agent, three layers, and two interfaces that have to hold

The decomposition everyone already uses, written down precisely enough to reason about



The synthesis problem, stated once

Find a policy whose closed-loop trace satisfies a temporal logic specification, under the dynamics and state and input constraints. Each layer solves a piece of it, at its own rate.

Why a reduced-order model in the middle

Full dynamics are intractable at any useful horizon. A model too coarse yields plans the controller cannot track. The middle layer is that bargain.

The interfaces carry the guarantee

Top: every planned point stays inside the abstract state the decision layer committed to. Bottom: the controller tracks the reference. If both hold, satisfaction at the top survives down to the torques.

IN CO-DESIGN TERMS each layer is a design problem. Its reference is a resource the layer below must provide, and consistency across an interface is a refinement check.

Layers are cheap to draw. The content of the theme is what has to be true where two of them meet.

Theme B. Objectives and tasks

TODAY	WITH SCHEMA
Loop rates, abstraction resolution, reduced-order models and tracking objectives are chosen by hand, by a good engineer, once.	The five artful decisions become synthesis knobs with a criterion attached, so the choice can be made again when the platform or mission changes.
A plan can be infeasible for the dynamics and nothing in the method says so until the robot falls over.	Consistency between layers measured with bisimulation metrics and trajectory lifting. The mismatch gets a number, and contracts between layers react to it.
Every platform gets its own stack, and legacy or black-box components cannot be integrated with any guarantee.	A library of reduced-order models and contract-mediated interfaces reused across quadruped, humanoid and UAV, with safe augmentation of black-box parts.

B.1 Theory of layered control architectures

Reduced-order models that bridge planning and control, a library per platform, consistency metrics, system-level synthesis for layer design.

Matni, Ames, Pappas, Sangiovanni-Vincentelli, Zardini, Jadbabaie

B.2 Operationalization

Models just expressive enough to admit feasible plans. Primal-dual compatibility between planning and control. Contracts between layers with fallback on violation.

Matni, Ames, Sangiovanni-Vincentelli, Zardini, Jadbabaie

B.3 Scalability

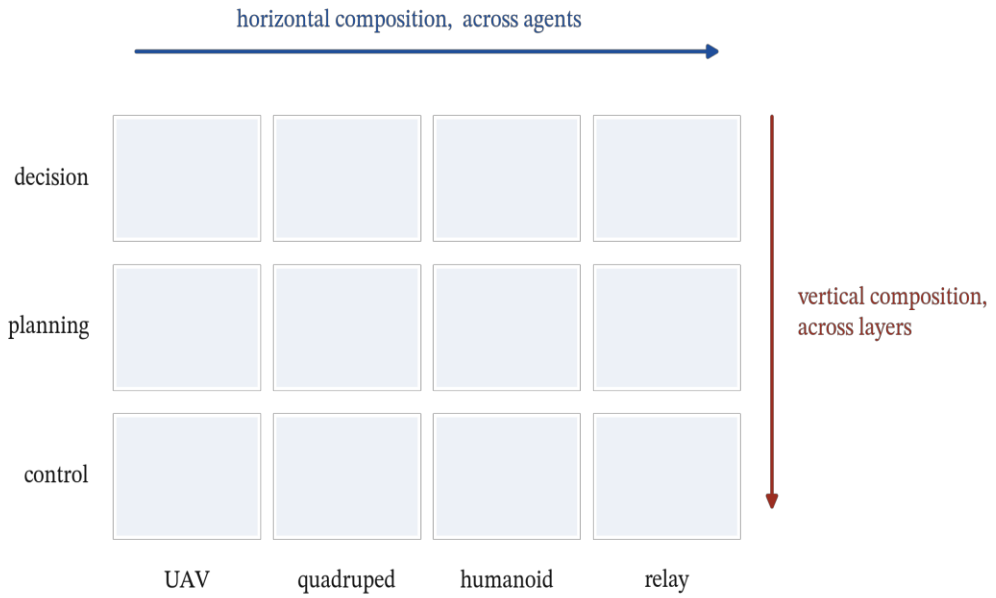
Online learning of residual dynamics. Safe augmentation of legacy and black-box components. Decomposition-based synthesis at fleet scale.

Matni, Jadbabaie, Ames, Zardini

Year 1 Task B.1 through the year. Deliverables, the reduced-order model library for quadruped, humanoid and quadrotor, the inter-layer consistency measures, and existing stacks adapted and instrumented on the G1, the Go2 and the custom drone.

Theme C. Horizontal and vertical are two axes of one problem

Multi-agent methods assume the stack inside each robot. Layered control rarely looks at the neighbors



Theme C treats both directions as dimensions of one design problem

A meta-model, not a bridge

Every controller, planner and communication link is a component with functionalities, resources and trade-offs. Horizontal composition is series and parallel of agents. Vertical composition is abstraction of dynamics into interfaces.

Compositional Pareto optimality

An architecture is a set of choices across all agents and all layers. It is Pareto optimal when no component and no layer can change to improve one system objective without worsening another.

When is diversity worth it

The sweet-spot measure compares the most balanced Pareto point to every other one. Small means one architecture is near optimal in every objective. The hypothesis, diversity across layers and agents shrinks it, but only for diverse tasks.

IN CO-DESIGN TERMS the wiring diagram is no longer given. Which boxes exist and how they connect becomes part of the implementation space, and the Pareto front is taken over architectures.

Adapting one sensing loop on one UAV should be able to trigger a task redistribution across the fleet, and be shown to be safe.

Theme C. Objectives and tasks

TODAY	WITH SCHEMA
Fleet-level methods take each robot's stack as given. Layered control takes the fleet as given. Nobody optimizes both at once.	▸ One co-design meta-model where agents and layers are two dimensions of the same problem, and the architecture itself is a design variable.
Architectures are compared by anecdote. There is no metric for whether a decomposition is good or how far a local change propagates.	▸ Compositional Pareto optimality, the sweet-spot measure, coupling sensitivity and multi-agent bisimulation metrics, computed rather than argued.
A local failure triggers either a global redesign or nothing at all.	▸ Contract-aware synthesis and runtime monitors that confine the correction to the affected part, with legacy components wrapped in bounding contracts.

C.1 Unified compositional and hierarchical decision-making

The meta-model. A formal definition of compositional Pareto optimality. Architecture metrics, sweet spots and coupling sensitivity via Markov categories.

Zardini, Ames, Sangiovanni-Vincentelli · Pappas, Jadbabaie

C.2 Operationalization

Contract-aware synthesis toolchain across agents and layers. Runtime monitor with localized correction. Contract wrapping for legacy parts. Barrier functions as safety enforcers.

Ames, Sangiovanni-Vincentelli · Matni, Zardini, Pappas

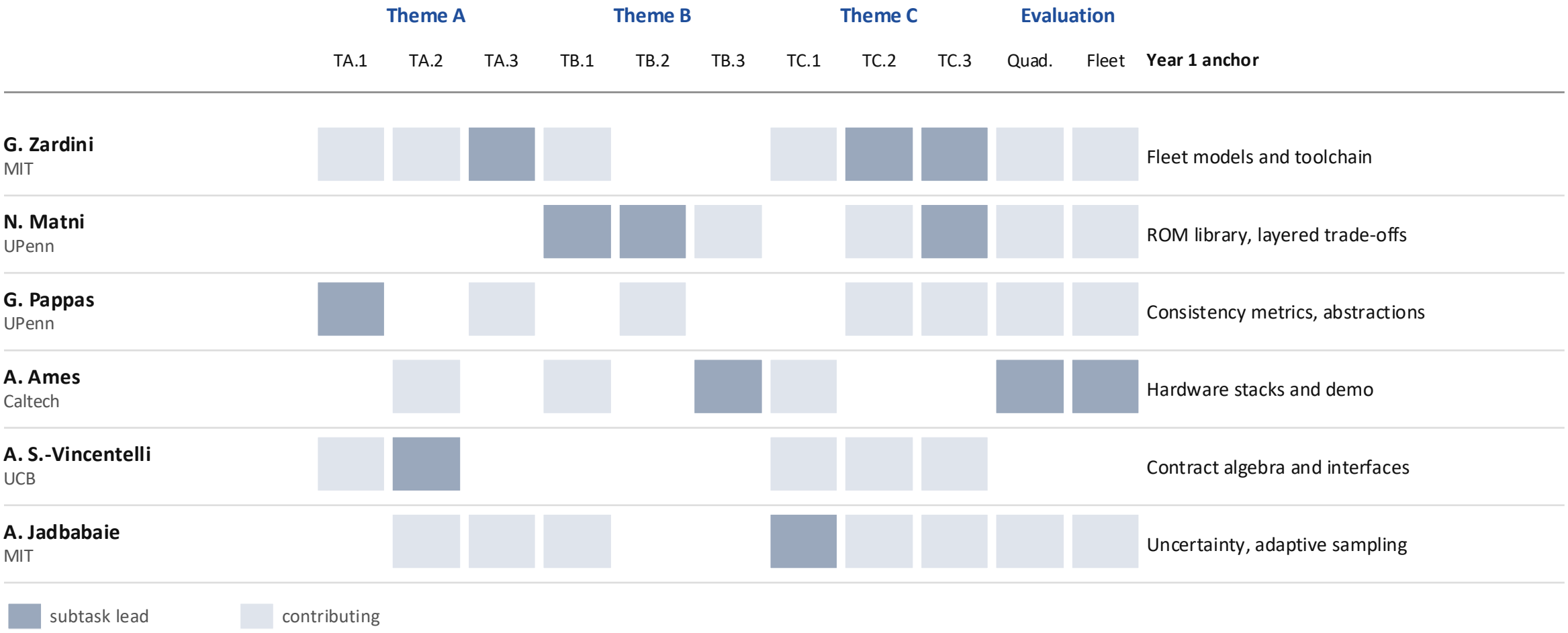
C.3 Scalability

Distributed optimization across agents and layers. Bayesian exploration of the design space. Federated learning. Learning-embedded verification. A runtime scheduler guided by the sweet-spot measure.

Zardini, Matni, Jadbabaie · Pappas, Sangiovanni-Vincentelli, Ames

Year 1 Task C.1 through the year. Deliverables, the meta-model with horizontal and vertical as two dimensions of one co-design problem, a first formal statement of compositional Pareto optimality, and one worked phase diagram with a layer choice on an axis.

Roles across themes and validation



Every task has at least two institutions on it.

Year 1, what we will have written down

Theme A

- Design-problem models of the mixed fleet. Perception, locomotion, comms, energy and infrastructure, each with a contract interface.
- The first uncertainty-aware version, with bounds and parametric distributions carried through composition.
- A working skeleton of the synthesis pipeline from specification to Pareto-optimal fleet.

Theme B

- A library of reduced-order models for quadruped, humanoid and quadrotor, each with the trajectory class it can deliver.
- Formal inter-layer consistency measures, bisimulation metrics and trajectory lifting, evaluated on real stacks.
- The five artful decisions mapped onto explicit synthesis knobs.

Theme C

- The meta-model. Horizontal and vertical as two dimensions of one co-design problem.
- A formal statement of compositional Pareto optimality and of the sweet-spot measure for layered multi-agent systems.
- One worked phase diagram with a layer choice on an axis.

Tasks A.1, B.1 and C.1 run through the whole of Year 1. Tasks A.2 and B.2 start in the second half. Everything above is a target, and we expect to be held to it next September.

By the end of Year 1 each theme has its core object defined and one worked example behind it.

Year 1, what we will have running

Three platforms that already fly and walk today, instrumented so their capability becomes an input to the design framework

Quadruped, Unitree Go2

layered architecture figure

single-integrator reduced-order model

Quadrotor, custom UAV

layered architecture figure

three-dof double-integrator model

Humanoid, Unitree G1

layered architecture figure

the same single-integrator model

One mid-level model inside both the quadruped and the humanoid. Year 1 measures how far that reuse goes, and where it stops.

- Adapt the existing layered stacks to the G1, Go2 and custom drone; establish the baseline vertical integration.
- Quantify capability (speed, endurance, sensing, payload, compute) so each robot enters Theme A as a populated design problem.
- Run contract-based co-design on the current toolchain; score compositional feasibility against mission-constrained optimality.
- Close the year with a combined outdoor search-and-rescue demo, Penn quadrupeds and quadrotors under VICON.

Caltech leads validation, supported by Penn and MIT, on hardware that exists today.

Milestones, by quarter

Q1

- Students and postdocs on board.
- Shared repositories and experimental infrastructure.
- Weekly meeting and monthly seminar begin.
- First AFRL coordination meeting.

Q2

- Fleet design-problem models, first version.
- Reduced-order model library, first version.
- Baseline stacks running on G1, Go2 and drone.

Q3

- First contract-based fleet co-design study on the search-and-rescue scenario.
- Meta-model draft and compositional Pareto definition circulated.

Q4

- Combined outdoor demonstration.
- First SCHEMA workshop, program officers invited.
- Public release of first models and datasets.

Quarters are relative to project start. Every item is a target, not a contractual date.

Dates are targets. The cadence on the next slide is what makes them realistic.

How we work together

Weekly

Team meeting

One hour, all PIs, postdocs and students across four institutions. Progress and blockers, then one deeper item agreed in advance.

Monthly

SCHEMA seminar

One student or postdoc presents forty minutes of work in progress. PIs, AFRL and DIGICAMS invited. A stage for students, a pressure test before submission.

Quarterly

Theme deep dives

Theme leads convene their task groups. Milestones reviewed and interfaces checked both ways: A to B specifications, B to A infeasibility, C metrics over both.

Always

Cross-pollination

Every student has a second advisor at another institution. Exchange visits, MIT and Penn to Caltech hardware, Caltech to MIT and Penn for theory. Joint sessions with DIGICAMS on shared objects.

Always

Shared infrastructure

One code organization, one benchmark scenario, one dataset format, so a result moves between labs without translation.

Yearly

In-person workshop

Rotating between PI institutions, program officers included, with a tutorial day on contracts, co-design tools and benchmarks for AFRL researchers.

Weekly to stay aligned, monthly to make the students visible, yearly to make it real.

How we want to work with AFRL

- Quarterly coordination meetings, on site or virtual, on technical alignment and on where transition opportunities are opening.
- Residencies and internships. AFRL researchers spend short periods at Penn and MIT on the theory.
- Caltech hosts AFRL students and early-career scientists for hands-on time on the robots.
- An annual SCHEMA tutorial on contract-based synthesis, the co-design tools and the benchmark platforms.
- Open-source repositories with reproducible results, so anything we claim can be checked and adapted.

A question for the breakout at 1400

Which mission scenarios, and which trade-offs, should the first phase diagrams be drawn for?

All six PIs have had AFOSR support from the Dynamical Systems and Control and Computational Mathematics programs, so these channels are already open.

Tell us today what belongs on the axes.

SCHEMA today

1135	Trade-offs in layered control architectures	Nikolai Matni, Penn
1205	Categorical Lyapunov	Aaron Ames, Caltech
1330	Compositional design of society-critical systems	Gioele Zardini, MIT
1400	Breakout discussion	Gioele Zardini with Matthew Klawonn, AFRL

SCHEMA is building the theory and the tools to draw phase diagrams for autonomy architectures, with guarantees that survive composition across agents and across layers.



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Thank you. Questions.